GreenTwin

System design: five parts, their contracts, data and build plan

Megathon X: Singularity · Lloyds Problem Statement 2: Green Home Score and Retrofit Recommendation Engine · Team MegaMind · contracts schema 0.4.0

> **In one line.** A Bayesian-calibrated physics twin of every home, which you can see in 3D, explain in plain language and act on, where every number is backed by physics and the range it comes with.

Audience: the build team. This document is the high-level design: the five parts, how they connect, the data that feeds them and the build plan. The companion **LLD.pdf** holds every contract field, interface signature, algorithm, table schema and download step. Binding rules: AGENTS.md and contracts/.

Section	What it covers
1. Philosophies	The six principles every part follows; how the brief's core rule is enforced
2. The five parts	What each part answers, what is inside it, coupled core vs independent parts
3. Architecture	End-to-end flow, the what-if loop, technology choices
4. Part designs	Purpose, components, models, contracts in/out and done-when for each part
5. Contracts and interfaces	The contract and Protocol each part owns (full detail in LLD.pdf)
6. Data	Datasets and how to get them, the EPC live tables, joins, synthetic data, day-1 plan
7. Uncertainty	Every number as a full distribution
8. Key mechanisms	Improvement ladder, improvement potential
9. Evaluation and submission	Metrics vs baselines, PS coverage, submission items
10. Build plan	Work breakdown, checkpoints, coding guidelines, open decisions

1. Core philosophies

#	Philosophy	What it means in the system
1	**Physics is the source of truth**	Every energy, savings, carbon and rating number comes from the Level-1 digital twin (Part 1). ML sets priors (EBM), estimates value and checks results; it never produces a customer's savings, so the system handles retrofit combinations no dataset has seen.
2	**Bayesian in thought**	Each home is calibrated on its own meter readings. Every output is a full distribution (mean, median, mode, std, percentiles, 90% interval, tail probabilities) plus P(Poor), P(suitable), P(on front). Missing data never becomes zero; it widens the range.
3	**Rerun, never add up**	A retrofit is a change to the twin's parameters; every what-if and package reruns the model. Interactions come from the physics. Savings are always per fuel.
4	**Grounded explanation**	The LLM only quotes tool outputs; a validator rejects any other number; Jev only routes questions and screens feasibility. No backing → "unavailable".
5	**Explainability first**	Score waterfall, heat loss per element, EBM contributions, value coefficients, a reason on every recommendation or veto.
6	**See it to believe it**	A living 3D twin: heat loss glows, retrofits appear, numbers update live, all driven by the same calibrated model.

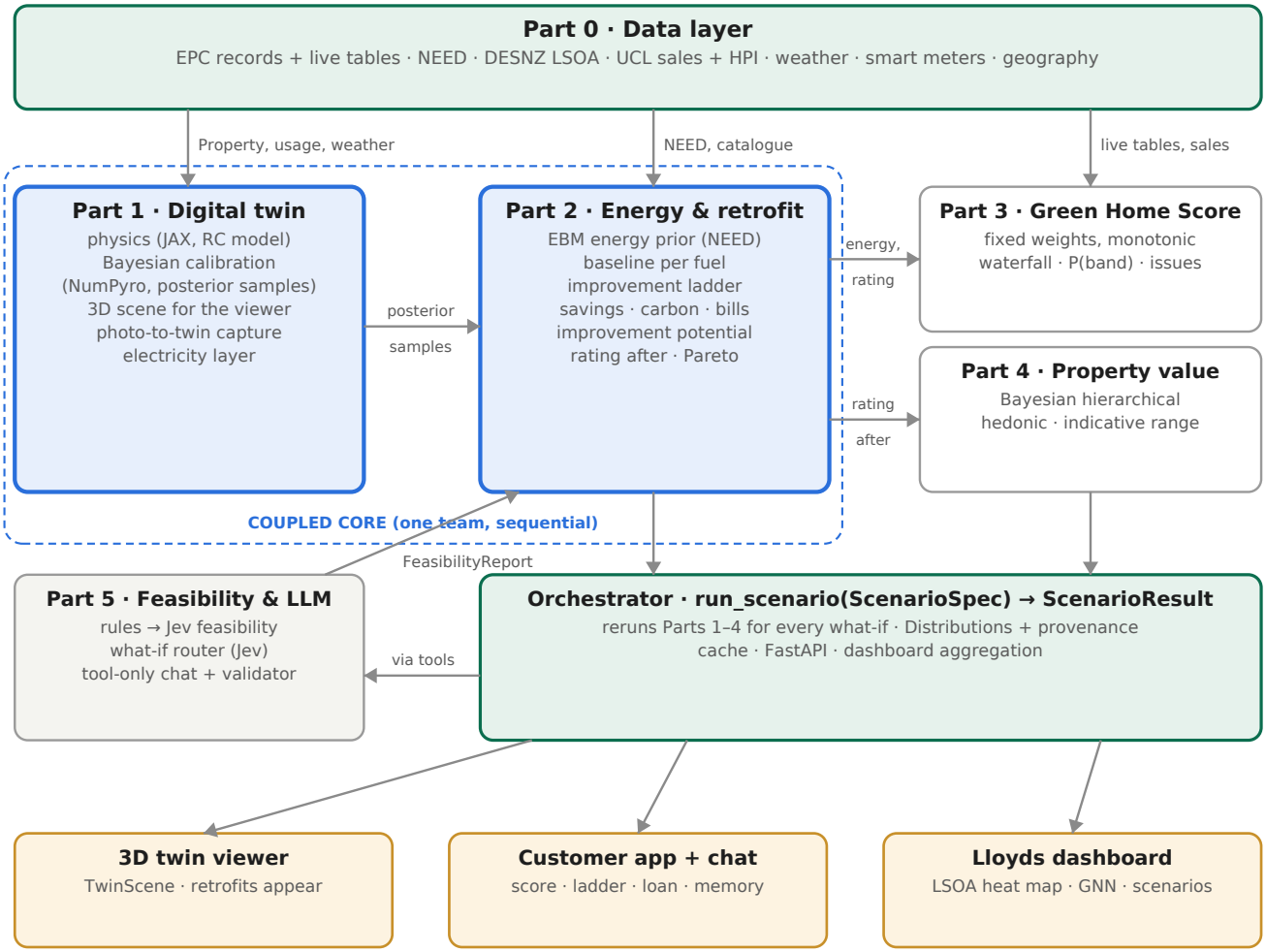
The brief's core rule, enforced by design

"*The score, savings, carbon and value figures come from documented models and calculations. The LLM only explains them... For a what-if request, the relevant models must first be rerun.*" (PS §1)

- Only Parts 1–4 produce numbers, each with provenance naming its model.
- Every what-if goes through **run_scenario** before the LLM sees anything.
- The LLM's answer passes a numeric validator → regenerate once → template explainer.

2. The five parts

GreenTwin is five parts on top of a shared data layer and a thin orchestrator. Parts 1 and 2 form the **coupled core**: every energy number runs through the twin, so they are built by one team in sequence. Parts 3, 4 and 5 (and the data layer and UI) are **independent**: each codes against its interface and a stub, and connects to the core at a few fixed points.



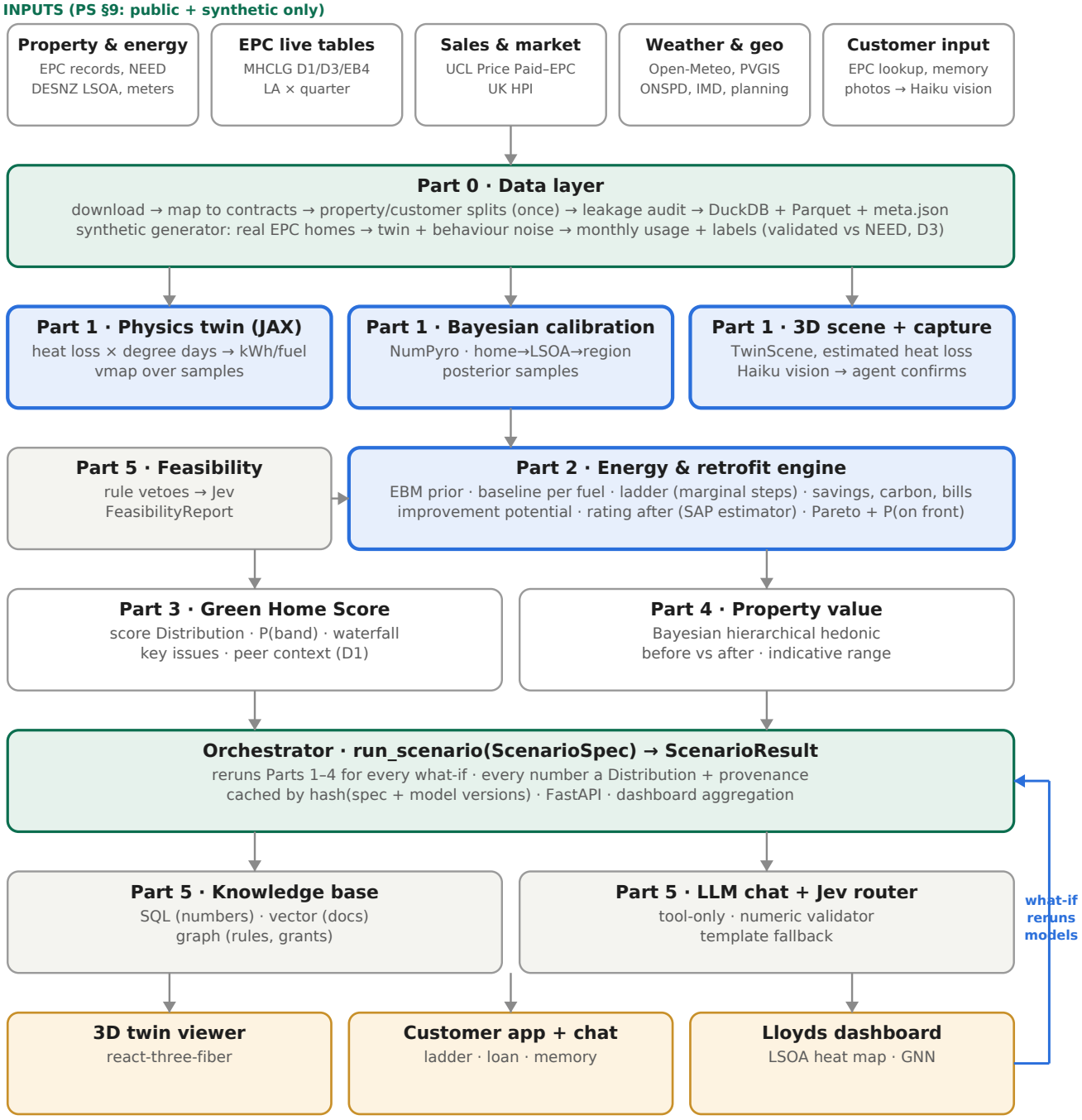
Blue = coupled core (every energy number runs through the twin). Green = data + orchestration. Grey = LLM side (reads only via tools).
Parts 0, 3, 4, 5 and the UI are independent: each codes against its interface (greentwin/interfaces.py) and a stub, then plugs in.

Part	Answers (PS)	Inside	Coupling
1 · Digital twin	the physics behind §3, §5, §6	JAX physics twin · NumPyro Bayesian calibration · 3D scene · photo-to-twin capture · electricity layer	Core
2 · Energy & retrofit	§3 potential, §4 retrofits, §5 savings, §6 carbon	EBM energy prior · baseline · improvement ladder · savings, carbon, bills · improvement potential · rating after · Pareto	Core
3 · Green Home Score	§2 score, §3 poor homes	fixed-weight score · sub-scores · waterfall · key issues · projected score · peer context	Independent
4 · Property value	§7 value	Bayesian hierarchical hedonic model · before vs after · green twins	Independent
5 · Feasibility & LLM	§1 core rule, §4 feasibility, §8 what-if, §10 memory	rules → Jev feasibility · what-if router · tool-only chat + validator · knowledge base · memory	Independent

The four connection points

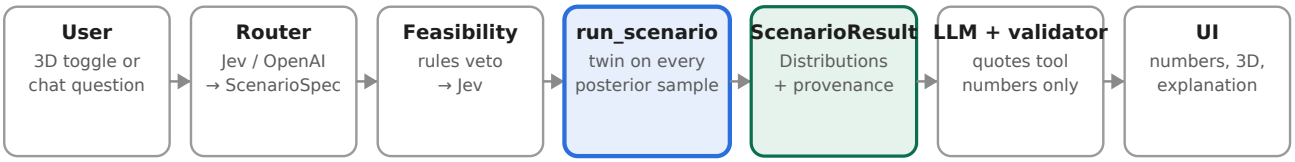
From → to	Contract	Until it exists, use
Part 2 EBM → Part 1 calibration	EnergyPrior (prior mean per fuel)	RdSAP defaults
Part 5 feasibility → Part 2 ladder	FeasibilityReport	rules only
Part 2 rating → Parts 3, 4	RatingEstimate (estimated SAP after)	stub: band + 1
Orchestrator → Part 5 LLM, UI	ScenarioResult, TwinScene	contracts/examples/*_MOCK.json

3. Architecture (end to end)



Every arrow is a contract in contracts/models.py; the experience layer never computes, it renders ScenarioResult / TwinScene.

The what-if loop



Low router confidence → clarifying question. Validator failure → regenerate once → template explainer. Missing data → "unavailable" + reason.

Technology choices

Concern	Choice
Data + storage	Python 3.12, pandas/Polars, DuckDB + Parquet with a meta.json per file
Physics twin	**JAX** (jax.numpy): runs inside the NumPyro model so NUTS/SVI get exact gradients; jax.vmap over posterior samples. Monthly steady-state RC model; differentiable RC graph (diffrax) later
Bayesian	NumPyro: NUTS offline for the hierarchy, SVI per home (< 1 s); ArviZ diagnostics
Electricity layer	Half-hourly: smart-meter load-shape archetypes + heat pump + PVGIS PV − battery
Learned models	EBM (InterpretML) energy prior, XGBoost baseline; GraphSAGE GNN (PyTorch Geometric) on the LSOA graph; Bayesian hedonic value model (NumPyro)
Photo-to-twin	Claude Haiku 5.5 (claude-haiku-5-5) vision + structured outputs; react-konva editor
LLM + router	OpenAI Responses API tool use (LLM_MODEL = gpt-6-luna) + numeric validator + template fallback; Jev System 1 if a key exists, else OpenAI structured output
API + front end	FastAPI; React + Vite + TypeScript, react-three-fiber + drei, Recharts, MapLibre + deck.gl, Zustand; types generated from the contracts

4. Part designs

Part 0 · Data layer

PS §9 · feeds every part

Purpose. Clean, licensed, split, joinable real data and every external factor traceable to a cited source.

- Download scripts write raw files untouched, then Parquet + meta.json (DatasetMeta) per source.
- EPC record → Property mapper: regex rules turn EPC text (walls, roof, glazing, heating, controls) into contract enums; ≥ 95% mapped per field; unmapped values logged.
- EPC live tables (MHCLG D1/D3/EB4) → AreaEPCStats per local authority × quarter: calibration targets, validation checks, peer context and dashboard layers (section 6).
- Splits: 70/15/15 by customer, else property, seed 42, assigned once; spatial-block splits for area models; leakage audit (epc.potential_* are labels only).
- Synthetic generator (uses the Part 1 twin): real EPC homes + NEED-calibrated behaviour → monthly usage with gaps and conflicts + reference labels; reweighted to EPC live-table band and type mixes; validated vs NEED.

Models (primary · baseline)	— (data engineering); EPC mapper, k-means load profiles
Consumes	public datasets, customer input
Produces	Property, EnergyUsageMonth, SaleRecord, WeatherMonth, AreaEPCStats, LSOAConsumption, LoadProfileArchetype, EPCRecommendation, ReferenceLabel, DatasetMeta
Done when	No id spans two splits; mapper coverage ≥ 95%; our EPC counts reconcile with live table D1; every table has meta.json; DATA section of the LLD complete.

Part 1 · Digital twin (physics, Bayesian, 3D)

PS §3, §5, §6 physics · coupled core

Purpose. A documented physics model of each home, calibrated on its own bills, that every energy number runs through, and that the customer can see in 3D.

- Physics (JAX): heat loss Σ U·A + ventilation × degree days → space heat; hot water; baseload; fuel via boiler efficiency or heat-pump SCOP; PV cuts imports. Calibrated and standard (SAP occupancy) modes.
- Bayesian calibration (NumPyro): heat-loss multiplier, setpoint, baseload, efficiency; hierarchy home → LSOA → region; Student-t likelihood on monthly meters; missing months latent; SVI per home, NUTS offline.
- 3D scene: parametric house (or captured rooms) with surfaces coloured by estimated heat-loss share and retrofit slots; never calculates.
- Photo-to-twin: Haiku vision drafts rooms + element tags with confidence; agent confirms in a react-konva editor; room geometry improves areas and the 3D house.
- Electricity layer: half-hourly profile for PV self-consumption, battery and time-of-use bills.

Models (primary · baseline)	Level-1 RC twin (physics) · Bayesian calibration (vs degree-day OLS, LightGBM quantile)
Consumes	Property, EnergyUsageMonth, WeatherMonth, EnergyPrior, LoadProfileArchetype
Produces	TwinParams, CalibrationResult, PosteriorSamples (in-process), ElementHeatLoss, TwinScene, CaptureResult, ElectricityProfileSummary
Done when	SH1–SH3 within ±15% of metered totals; monotone sweep passes; 90% coverage ≈ 90% on held-out homes; standard-mode kWh/m² tracks EPC live table D3 by local authority.

Part 2 · Energy calculation & retrofit efficiency

PS §3, §4, §5, §6 · coupled core

Purpose. Turn the calibrated twin into the numbers the brief asks for: baseline energy per fuel, what each retrofit adds, savings, carbon, bills, improvement potential, the estimated rating after, and the best packages.

- Energy prior: EBM on NEED predicts kWh/m² per fuel (prior for homes without meters) with feature contributions.
- Baseline: per-fuel Distributions; intensity = annualised metered energy per m² (the brief's definition).
- Improvement ladder: feasible retrofits in fabric-first order; each step's marginal effect on every posterior sample; P(suitable) = P(feasible) × P(net benefit); probabilistic verdicts.
- Improvement potential: all feasible retrofits in one run; null + reason if the baseline is 0.
- Rating after: SAP 2012 formula on the twin's standard mode, anchored to the recorded SAP.
- Pareto: enumerate feasible packages; P(on front) across samples; knee.

Models (primary · baseline)	EBM energy prior (vs NEED median, XGBoost) · SAP estimator (vs median band shift) · Pareto (vs greedy kWh/£)
Consumes	PosteriorSamples, FeasibilityReport, catalogue, emission factors, tariffs
Produces	EnergyPrior, EnergyBreakdown, RetrofitEffect, ImprovementPotential, RatingEstimate, ParetoPoint
Done when	SH2 shows electricity rising after a heat pump; package = one run; potential null at baseline 0; rating change validated on EPC before/after pairs; precision@k vs EPC assessor recommendations.

Part 3 · Green Home Score

PS §2, §3 · independent

Purpose. Answer "how green is my home" with the brief's fixed weights, a transparent waterfall and the issues that matter.

- S = Σ w·s from contracts/score_mapping.json; seven monotonic, documented sub-scores; rounded; five bands.
- Energy efficiency from the recorded SAP (or the estimated rating for a projected score); intensity from annualised metered kWh/m².
- Score Distribution and P(each band) from posterior samples; key issues with evidence; peer context from the EPC live tables (local-authority band shares, D1).

Models (primary · baseline)	Deterministic scoring (no training)
Consumes	Property, EnergyBreakdown, RatingEstimate, AreaEPCStats
Produces	ScoreResult (components, KeyIssue list, band probabilities)
Done when	Weights sum to 1; monotonicity sweep; every heating type mapped; projected ≥ current for improvement-only packages.

Part 4 · Property value

PS §7 · independent

Purpose. An indicative value range, before vs after, accounting for location, size, type and market.

- Bayesian hierarchical hedonic regression on real UK sales: log(price ÷ HPI) = pooled location effects (LSOA within LA within region) + size + type + age + tenure + β(region × type) × SAP.
- Same home at its recorded vs estimated-after rating, applied to the current value estimate; full distribution; wording always "indicative"; range including 0 → no clear effect.
- Green twins: matched greener comparable homes shown as ranges.

Models (primary · baseline)	Bayesian hedonic (vs OLS without the rating)
Consumes	Property, SaleRecord (training), RatingEstimate
Produces	ValueImpact, GreenTwin
Done when	Held-out coverage and pinball loss beat the baseline; β compared with Nationwide / DECC 2013; planted-premium recovery reported as a method check only.

Part 5 · Feasibility rules engine, Jev, LLM explanation and chat

PS §1, §4, §8, §10 · independent

Purpose. Decide what is possible for a home, route what-if questions, and explain computed numbers without ever inventing one.

- Feasibility: rule vetoes from contracts/feasibility_rules.json always win; Jev classifies survivors with recalibrated confidence; low confidence → needs survey; OpenAI structured output without a Jev key.
- Router: Jev intent + ScenarioSpec slots; below the threshold → clarifying question.
- Chat: tool-only gpt-6-luna (get_home_facts, get_energy_prior, run_scenario, rank_recommendations, search_docs, query_graph, memory tools), numeric validator, regenerate once, template fallback; knowledge base (SQL, vector, graph); customer memory.

Models (primary · baseline)	Jev threshold calibration (isotonic) · LLM evaluated on ≥ 30 adversarial scenarios
Consumes	Property, catalogue, CustomerMemory, ScenarioResult (via tools)
Produces	FeasibilityReport, WhatIfRoute, ChatResponse, ExplanationCheck, CustomerMemory
Done when	Rule veto beats Jev; 0 unvalidated numbers; groundedness and accuracy ≥ 90%, unsupported claims < 5%; works with no keys.

5. Contracts and interfaces

Parts exchange data only through Pydantic contracts (contracts/models.py, schema 0.4.0; TypeScript mirror generated into contracts/types.ts) and call each other only through the Protocols in greentwin/interfaces.py. Every field and signature is listed in LLD.pdf, generated from the code.

Part	Interface (Protocol)	Key methods	Owns these contracts
0 Data	DataLayer	load_property, load_usage, load_weather, load_sales, load_area_epc_stats, load_lsoa_consumption, load_load_profiles, catalogue	Property, EnergyUsageMonth, SaleRecord, WeatherMonth, AreaEPCStats, LSOAConsumption, LoadProfileArchetype, EPCRecommendation, ReferenceLabel, DatasetMeta
1 Twin	DigitalTwin	params_from_property, apply_retrofits, simulate, calibrate, element_heat_loss, electricity_profile, scene, capture	TwinParams, CalibrationResult, ElementHeatLoss, TwinScene, SceneSurface, RetrofitSlot, FloorPlan, Room, ElementTag, CaptureResult
2 Energy	EnergyRetrofitEngine	energy_prior, baseline, ladder, package, improvement_potential, estimate_rating, pareto	EnergyPrior, EnergyBreakdown, MonthlyEnergy, ElectricityProfileSummary, RetrofitEffect, ImprovementPotential, RatingEstimate, ParetoPoint
3 Score	GreenHomeScorer	score	ScoreResult, ScoreComponent, KeyIssue
4 Value	ValueModel	value_impact, green_twins	ValueImpact, GreenTwin
5 Feasibility & LLM	FeasibilityEngine, Router, Explainer	assess, route, chat	Feasibility, FeasibilityReport, WhatIfRoute, ChatRequest, ChatResponse, ExplanationCheck, Citation, CustomerMemory
Orchestrator	Orchestrator	run_scenario, dashboard_areas	ScenarioSpec, ScenarioResult, DashboardArea
Shared	contracts.summarise	summarise, exact, category_probabilities	Distribution, Provenance, FeatureContribution

Rule for every part: uncertain numbers are Distributions built with contracts.summarise; every result carries Provenance; missing data is None (never 0); savings are per fuel; packages are re-run, never summed. A test fails if types.ts drifts from models.py.

6. Data

No public dataset links property features, monthly meter readings and sale prices for the same homes. So: **real data for every part that exists publicly**, joined at the level each dataset allows, plus **physics-generated synthetic data only for the missing piece** (monthly readings per home and the labels), calibrated and validated against real aggregates.

Datasets and how to get them

Dataset	Used by	Access	Licence
EPC domestic certificates + recommendations	Part 0 homes · Part 3 score inputs · Part 2 rating check (certificate history) · Part 5 rules · recommendation labels	Bulk CSV download + developer API (OpenAPI 3, bearer token)	OGL v3 except where stated (see the service's licensing restrictions page)
EPC live tables (statistics)	Part 0 calibration targets + download check · Part 1 twin validation · Part 3 peer context · dashboard	Direct ODS download, quarterly (data to June 2026, published 30 July 2026)	OGL v3
NEED anonymised data 2025	Part 2 EBM energy prior (training) · Part 0 synthetic behaviour calibration and validation	Direct download: 50,000-row and 4,000,000-row property-level samples + metadata	OGL v3
Sub-national domestic electricity and gas by LSOA	GNN area-model labels · dashboard observed layer · synthetic area check	Direct XLSX download (LSOA domestic electricity, LSOA domestic gas)	OGL v3
UCL linked Price Paid + EPC, 1995–2024 (ReShare 857911)	Part 4 Bayesian hedonic value model (training) · green twins	Open download, no registration: 857911_bundle_1.zip (3 GB) + bundle_2.zip (4 GB)	CC BY 4.0
HM Land Registry Price Paid Data	Part 4 recent sales after Oct 2024 (needs linking to EPC)	Direct CSV (complete file, yearly, monthly)	OGL v3 with attribution (address data has extra rights notes)
UK House Price Index	Part 4 market adjustment (deflate prices to today)	CSV download	OGL v3
Smart meter energy consumption, London households (Low Carbon London)	Part 1 electricity layer: half-hourly load-shape archetypes	Direct download (combined CSV ≈ 10 GB unzipped, ~167M rows; tariff file)	Open (see the London Datastore page)
Open-Meteo Historical Weather API (ERA5)	Part 0/1 monthly heating degree days and irradiance per postcode district	REST GET, JSON; cache every response	Data CC BY 4.0; cite Open-Meteo and ERA5 with the Copernicus acknowledgement
Met Office HadUK-Grid	Documented degree days and 30–50-year regional trends (dashboard)	CEDA archive download	OGL
PVGIS 5.3 API	Part 1 PV yield (monthly PVcalc, hourly seriescalc)	REST GET at https://re.jrc.ec.europa.eu/api/v5_3/{PVcalc|seriescalc|tmy}	Free use with attribution to EU JRC PVGIS

Dataset	Used by	Access	Licence
ONS Postcode Directory (ONSPD)	The join key: postcode → LSOA 2021, local authority, region, lat/lon	CSV download (May 2026 release; August 2026 user guide published)	OGL (contains OS and Royal Mail data; attribution required)
English Indices of Deprivation 2025	Features for the EBM prior, value model and GNN; dashboard	Direct download (File 1 IMD; download-all tool at deprivation.communities.gov.uk)	OGL v3
LSOA 2021 boundaries	Dashboard maps; LSOA adjacency graph for the GNN	GeoJSON / GeoPackage download	OGL
Conservation areas + listed buildings	Part 5 feasibility rules (EWI, glazing restrictions)	Direct CSV / GeoJSON / Parquet	OGL v3 (attribution: Historic England, Ordnance Survey)
Off-gas-grid indicator	Part 5 rules and grants (BUS off-gas uplift); dashboard layer	EPC field per home; postcode lists as XLSX/CSV	EPC: OGL; postcode lists: check terms (CSE 2017 listing has no licence)
Electrification of Heat heat-pump performance (SN 9050 / 9210 / 9209)	Part 1 heat-pump SCOP priors by house type	SN 9050 cleansed data: open, summary dataset downloadable; full cleansed data by FTP order	UK Data Service terms (check per study)
English Housing Survey energy tables	Part 0 stock distributions to fill gaps in the synthetic population	Direct XLSX tables	OGL v3
CubiCasa5k floor plans	Part 1 photo-to-twin evaluation (room polygons, areas)	Zenodo download	Check the repository LICENSE; treat as non-commercial until confirmed
Octopus Energy public products API (time-of-use tariff citation)	Part 2 time-of-use bill figures (cite one published tariff)	REST GET https://api.octopus.energy/v1/products/ ; product details list tariff unit-rate links	Octopus API terms; cite tariff, URL and access date in tariffs.json

Start here (day 1, in this order)

1. Register GOV.UK One Login and get EPC bulk access (the slowest step); use cached samples meanwhile.
2. Download the EPC live tables D1, D3, EB1, EB3, EB4 (ODS, minutes) and build area_epc_stats.
3. Download NEED 2025 (50k now, 4M for training) with its metadata.
4. Start the UCL 857911 bundles (≈ 7 GB) and the UK HPI CSV.
5. Download ONSPD (latest), IMD 2025 and LSOA 2021 boundaries: the join keys for everything else.
6. Download DESNZ LSOA domestic electricity and gas 2010–2024.
7. Fetch Open-Meteo weather and PVGIS yields per postcode district (cache every response).
8. Download conservation-area and listed-building files from planning.data.gov.uk.
9. Later: London smart meters (10 GB, stream with DuckDB), Electrification of Heat SN 9050 summary, CubiCasa5k, one Octopus time-of-use tariff for tariffs.json.

The EPC live tables (MHCLG): how they are used

The GOV.UK *Live tables on Energy Performance of Buildings Certificates* are quarterly ODS tables of EPC **lodgements** for England and Wales by local authority × quarter (D1: rating band counts; D3: floor area, modelled energy kWh/m², CO2, fuel cost; EB4: property type; EB-tables for existing dwellings only). They are aggregates, not homes, so they are used to anchor and check the per-home data, not to train per-home models:

Use	Table	How
Population calibration targets	D1, EB4	Rake our EPC sample and the synthetic population so band mix, property-type mix and floor area per local authority match the official totals for the same period.
Download / mapper check	D1	Our certificate counts per LA × quarter × band must reconcile with D1; a gap > 2% flags missing files, opt-outs or mapping errors.
Twin validation	D3	The twin's standard mode (SAP-like occupancy) aggregated per LA vs D3 mean modelled kWh/m² and fuel cost: an official check on the physics.
Score peer context	D1, D3	"Your home vs your area": local-authority band shares and median modelled kWh/m² in ScoreResult.peer_context.
Synthetic-data validation	D1, D3, EB4	Band, type and kWh/m² distributions of the synthetic homes vs the official tables per LA (chi-square, KS), reported in the evaluation.
Dashboard layer	D1, D3	Official LA layer beside our modelled LSOA layer: share A–C vs D–G, trend over quarters, mean modelled kWh/m², CO2 per dwelling.

Caveats handled: counts are lodgements in a period (repeat certificates, opt-outs), not the dwelling stock; a Q1 2026 local-authority allocation error was revised in May 2026; D3 averages were corrected in December 2023 (recompute averages from totals as the table notes explain).

How the data joins

Level	Join	Datasets
Postcode / LSOA	ONS postcode directory → LSOA 2021, LA code, region	EPC records, UCL sales, weather (by district), planning, off-gas, IMD, DESNZ LSOA
Local authority × quarter	LA code	EPC live tables ↔ our EPC sample, synthetic population, twin standard-mode output, dashboard
Archetype	type × age band × floor-area band × region	NEED (no postcode) ↔ EPC homes, EBM training, synthetic validation
Per home (synthetic)	real EPC home + real weather → twin + behaviour noise	monthly usage + labels

7. Uncertainty: every number is a distribution

Bayesian inference happens in two places: calibrating each home's physics on its bills (Part 1) and the value model on real sales (Part 4). Everything downstream **propagates** those posteriors: each output is computed once per posterior sample, then summarised with contracts.summarise. Intervals are never added together; the samples carry the correlations.

Part	Fields	Why
Centre	mean · median · mode	They differ when the result is lopsided
Spread	standard deviation · IQR	How uncertain
Percentiles + intervals	P1–P99 · 50% and 90% credible intervals	Full shape; the 90% interval is the headline
Tails	labelled tail probabilities · mean of the worst 5% · skewness	What could go wrong, and how badly
Shape + meta	histogram · unit · n_samples · headline (mean or median)	Density plot; traceability

Mean	Median	Mode	Std	P5	P95	P(saving > 0)	P(rise > £100)
−£35	−£42	−£55	£88	−£180	+£110	0.38	0.23

Example (illustrative): heat-pump bill change on SH2. The mean hides that about 1 in 4 homes like this could see the bill rise by over £100/yr while carbon falls with near certainty.

8. Key mechanisms

Improvement ladder: what each retrofit adds

Rules + Jev decide what is possible; the twin decides what it is worth. Step k's effect = twin(steps 1..k) − twin(steps 1..k−1) on every sample, fabric first. The steps add up exactly to the package (one run). Verdicts are probabilistic: Recommended if P(suitable) ≥ 0.8 and P(step adds ≥ 3% of purchased energy or ≥ 2 points) ≥ 0.8. Thresholds to be agreed.

#	Add (SH2, illustrative)	Gas kWh/yr	Elec kWh/yr	Potential	Score	CO2 t	Bill £	Verdict
1	Cavity wall insulation	2,600	0	17%	66	0.48	+207	Recommended
2	Loft top-up to 270 mm	900	0	23%	68	0.16	+72	Recommended
3	Draught proofing	400	0	26%	69	0.07	+32	Optional
4	Smart thermostat + TRVs	500	0	29%	71	0.09	+40	Recommended
5	Replace old double glazing	450	0	32%	72	0.08	+36	Not worth adding
6	Air-source heat pump	7,150	−2,300	64%	80	0.86	−35	Depends on goal
7	Solar PV 3 kWp	0	1,100	71%	86	0.22	+290	Recommended
	Package (one run)	**12,000**	**−1,200**	**71%**	**86**	**1.96**	**+641**	

Improvement potential (PS §3)

- 100 × (E_baseline − E_after) / E_baseline on purchased energy (grid import + gas), E_after from all feasible retrofits run together; a distribution with the per-fuel split.
- Needs-survey items reported separately; baseline 0 or missing → unavailable + reason; homes ranked by posterior mean with P(top 10%).

9. Evaluation and submission

Output	Metric (held-out, property splits)	Baseline	Part
Energy prior	MAE/MAPE per fuel, 90% coverage	NEED median, XGBoost	2
Calibration	coverage ≈ 90%, CRPS, R-hat; standard mode vs EPC D3 by LA	degree-day OLS	1
Rating after	MAE of SAP change on EPC before/after pairs	median band shift	2
Recommendations	precision@k vs EPC assessor lists; 0 infeasible	recommend all feasible	2, 5
Improvement ranking	Spearman / NDCG vs labels	rank by total kWh	2
Score	0 monotonicity violations; band-probability calibration	—	3
Value	coverage, pinball loss; β vs published	OLS without rating	4
GNN area model	MAE on held-out LSOAs (spatial blocks)	EBM without graph	0/dashboard
Photo capture	room-area error, tag accuracy, % fields corrected	manual tagging	1
LLM	groundedness ≥ 90%, accuracy ≥ 90%, unsupported < 5%	no validator	5
Synthetic data	KS vs NEED; band/type mix vs EPC live tables	—	0

PS	Requirement	Part
§1	Core rule; what-if reruns models first	5 + orchestrator
§2	Score with fixed weights, documented monotonic sub-scores, contributions	3
§3	Poor homes, key issues, improvement potential per fuel	3 + 2
§4	Feasible, prioritised retrofits with suitability, saving, reason	5 + 2
§5 · §6	Per-fuel savings with interactions; carbon from documented factors	1 + 2
§7	Indicative value range, before vs after	4 (+ 2 rating)
§9	Data, splits, leakage, documented synthetic labels	0
§10	Customer memory changes recommendations	5
§13	A prototype · B code · C models · D score doc · E LLM doc · F this document · G eval · H RAI · I test predictions	all

10. Build plan

Work breakdown

Track	Sequence	Starts on
Coupled core (Parts 1 + 2, one team)	physics twin (JAX) → rating estimator → synthetic generator → Bayesian calibration → ladder, potential, Pareto → electricity layer	RdSAP defaults, SH1–SH3 examples
Part 0 Data	EPC access + downloads → mapper → splits → DuckDB → live tables + DESNZ + smart meters	day 1
Part 3 Score	score + monotonic tests → key issues → peer context → projected score	recorded inputs
Part 4 Value	UCL sales prep → Bayesian hedonic → before/after → green twins	UCL sample; stub rating
Part 5 Feasibility & LLM	rules engine → mock server → chat + validator → eval set → Jev → memory	mocks
Experience	app shell on mocks → 3D twin v1 → capture editor → dashboard	mocks, generated types

Checkpoint	Done when
1. End-to-end skeleton	SH1–SH3 flow through the real twin → score → retrofits → API → UI
2. Complete submission	Bayesian distributions, rating estimator, value, grounded LLM, evaluation, docs
3. Demo build (freeze)	3D twin + capture, Pareto + loan, Lloyds dashboard, memory

Working rules (hackathon: speed first)

- Speed beats polish: thinnest end-to-end path first, mocks from day one, reuse and hardcode for the demo, timebox at ~45 minutes, commit small and often.
- Tests only where judges look: contracts, monotonicity, per-fuel signs, null potential at baseline 0, the LLM validator, no mock in the demo, types.ts in sync.
- Non-negotiables: contracts in and out; provenance; Distributions via summarise; missing ≠ 0; rerun never sum; the LLM never makes a number; value always indicative; no keys in git.
- One UI look: shared theme and components (NumberWithRange, DistributionCard, ScoreWaterfall), brand green #0b6e4f / blue #2a6fdb, colour-blind-safe scales, every number with unit + range.

Open decisions

Decision	Options	Recommendation
Twin ambition (library: JAX)	A: monthly steady-state first · B: differentiable RC graph (diffrax) first	A for checkpoint 1; upgrade behind simulate()
Verdict thresholds	gain threshold, probability cut-offs, cost per kWh	Agree once; record in SCORE doc
Default goal	bills or carbon	Bills by default; carbon toggle
Time-of-use tariff source	which published tariff to cite	Pick one public tariff; until then null